HUMAN GENETICS

Allele frequency selection and no age-related increase in human oocyte mitochondrial mutations

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Mitochondria, cellular powerhouses, harbor DNA [mitochondrial DNA (mtDNA)] inherited from the mothers. mtDNA mutations can cause diseases, yet whether they increase with age in human oocytes remains understudied. Here, using highly accurate duplex sequencing, we detected de novo mutations in single oocytes, blood, and saliva in women 20 to 42 years of age. We found that, with age, mutations increased in blood and saliva but not in oocytes. In oocytes, mutations with high allele frequencies were less prevalent in coding than noncoding regions, whereas mutations with low allele frequencies were more uniformly distributed along the mtDNA, suggesting frequency-dependent purifying selection. Thus, mtDNA in human oocytes is protected against accumulation of mutations with aging and having functional consequences. These findings are particularly timely as humans tend to reproduce later in life.

INTRODUCTION

Mitochondria are organelles that have a multitude of functions central to the eukaryotic cell, including the production of adenosine triphosphate (ATP) via oxidative phosphorylation, response to intrinsic and extrinsic stress by inducing metabolic signaling, etc. [reviewed in (*1*)]. Mitochondrial dysfunction manifests itself in diverse and frequently multisystemic diseases (*1*). How mitochondrial diseases originate is of great interest to the biomedical community. Mitochondrial dysfunction and subsequent diseases are caused in part by mutations in the mitochondrial genome, i.e., mitochondrial DNA (mtDNA) [reviewed in (*2*–*4*)], a 16.6-kb-long circular double-stranded DNA with 37 genes encoding 13 proteins, 22 transfer RNAs (tRNAs), and 2 ribosomal RNAs (rRNAs) [reviewed in (*5*)]. Many other mitochondrial genes are located in the nuclear genome (*6*). In contrast to nuclear DNA, with each chromosome usually present in two copies in somatic cells and one copy in germline cells, mtDNA is present in hundreds to thousands of copies in somatic cells [e.g., a median mtDNA copy number of 127 was measured in whole blood (*7*)] and >100,000 copies in mature oocytes [reviewed in (*8*)]. Human mtDNA is thought to be inherited exclusively through the maternal lineage (*9*).

Examining whether germline mutation frequency increases with age is highly relevant to modern societies in which reproduction later in life has become common. Whereas parental age has been linked to an increase in the frequency of germline mutations in the nucleus (*10*–*15*), the effect of maternal age on germline mutations in mtDNA has been debated. Pedigree studies have suggested an increased mtDNA mutation frequency in children born to older mothers [e.g., (*16*–*18*)]; however, post–fertilization selection might have affected them. Without the direct analysis of oocytes, we cannot ascertain whether an increased frequency of de novo mutations in the offspring is due to enhanced mutagenesis and selection in oocytes in older mothers versus due to post–fertilization selection in children born to older mothers. Whereas mtDNA mutations have been analyzed in human somatic tissues in detail [e.g., (*19*)], the direct examination of mtDNA mutations in human oocytes has been challenging due to methodological limitations. Most previous studies either focused on particular mtDNA sites [e.g., (*20*, *21*)] or used sequencing methods with high error rates [e.g., (*22*–*24*)]. Using a low-error duplex sequencing approach (*25*), we have recently shown that mutations across the whole mtDNA increase with age in mouse oocytes (*26*). Using the same approach, we have demonstrated that mtDNA mutations increase in macaque oocytes until ~9 years of age and do not increase afterward (*27*). We still do not know definitively whether the frequency of de novo mtDNA mutations increases with age in human oocytes.

To address this knowledge gap, we analyzed mtDNA substitution mutations in single oocytes, blood, and saliva from women of different ages. We used the highly accurate duplex sequencing method (*25*), which we had previously modified to generate high-quality mtDNA sequences directly from single oocytes (*26*, *27*). We obtained a comprehensive set of mutations to study the impact of age on frequencies of germline and somatic mutations, as well as on their distribution across mtDNA. We identified genetic variant hotspots and analyzed the effect of selection on the mutational landscape. In addition, we separately analyzed inheritable heteroplasmies—variants present in at least one oocyte and all somatic tissues assayed and therefore likely already segregating in the germ line during oogenesis. This allowed us to estimate the size of the effective germline bottleneck occurring in the oocyte lineage before fertilization and embryogenesis. Thus, we present a direct analysis of mtDNA variants in the human female germ line, unraveling their dynamics depending on age and allele frequency.

RESULTS

Duplex sequencing and mutation detection

To study the role of age in germline and somatic mtDNA mutation accumulation in humans, we generated high-quality, full-length mtDNA sequences for 22 women (mainly born in Europe) ranging

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in age from 20 to 42 years (Fig. 1A and table S1). For every woman, we assayed mtDNA from one to five single oocytes (mean = 3.64; SD = 1.05; a total of 80 single oocytes were assayed), blood ($n = 18$), and/or saliva ($n = 8$; Table 1 and table S1).

To identify DNA variants, we used duplex sequencing (*25*). By barcoding each double-stranded sequencing template before DNA amplification during library preparation, duplex sequencing separates true DNA variants from polymerase chain reaction (PCR) errors arising during library preparation and from sequencing errors. This is achieved by first forming consensus sequences for reads originating from each of the two template strands separately, i.e., single-strand consensus sequences (SSCSs), which still contain some PCR errors and artifacts resulting from DNA lesions, and next by forming a consensus of the two SSCSs, i.e., duplex consensus sequence (DCS), where the errors and artifacts are eliminated. We followed the protocol we previously developed for enriched mtDNA in mice (*26*). Briefly, the samples were enriched for circular mtDNA with the exonuclease V digestion of linear nuclear DNA (fig. S1A and table S2). The remaining DNA was sheared, sequenced with short [250–nucleotide (nt)] paired-end reads, and analyzed with the Du Novo software (*28*, *29*) to form DCSs (see Materials and Methods for details). We achieved a median mtDNA enrichment of 85.4, 0.7, and 4.0% and a median DCS depth of 1440×, 78×, and 158×, per site per sample for single oocytes, blood, and saliva, respectively (fig. S1, A and B, and table S2), with a rather uniform DCS depth across the mtDNA length (fig. S1C). The DCSs were mapped to the human reference genome including nuclear DNA and mtDNA, and only alignments with the best match to the mtDNA were retained (see Materials and Methods for details). At 219 different mtDNA positions, we identified fixed differences from the reference (supported

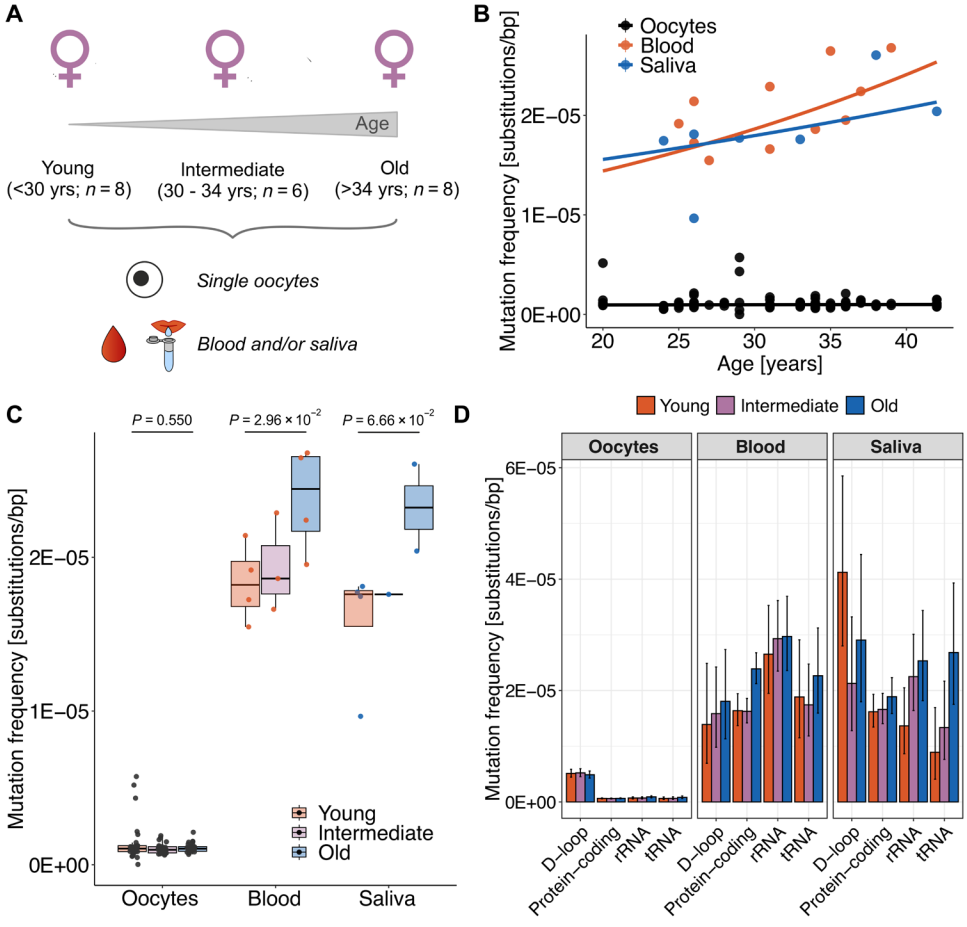


Fig. 1. mtDNA mutation frequencies in oocytes, blood, and saliva. (**A**) Oocytes, blood, and saliva samples were sequenced from 20- to 42-year-old women, assigned to three age groups: young, intermediate, and old. yrs, years. (**B**) Mixed-effects linear model analyzing the effect of age on mutation frequencies. Curves show the predicted mutation frequency based on the fixed-effect part of the model. Dots are the observed mutation frequencies per individual oocyte for oocytes and per tissue per donor for somatic tissues. (**C**) Mutation frequencies measured in single oocytes, blood, and saliva are shown for each donor and each age group. *P* values are from a permutation test comparing young and old age groups (one-sided test based on medians; 10,000 permutations; corrected for multiple testing; see Materials and Methods). Note that the three single oocytes showing higher mutation frequencies in (B) and (C) had a mean DCS depth of only 70×, 95×, and 117×, respectively; hence, their high mutation frequencies are likely an effect of the low number of mtDNA nucleotides assessed. Only 24 mutations (all measured in oocytes of young donors) are attributed to these samples. When removing these oocytes from the analysis, we still observed no significant differences between young and old women (median mutation frequencies of 9.46×10^{-7} and 1.02×10^{-6} substitutions/bp in young and old women, respectively; one-sided permutation test for equality in medians $P = 0.395$). (**D**) Tissue-specific mutation frequencies (i.e., the total number of tissue-specific mutations divided by the product of mtDNA region length and average DCS depth) for each age group in the D-loop (1122 bp), protein-coding regions (11,341 bp), rRNA (2513 bp), and tRNA (1505 bp). Mutation frequencies for noncoding sequences outside of the D-loop (88 bp) are shown in table S6. Mutation frequency bars are shown with 95% Poisson CIs.

Table 1. Donors and samples assessed with mtDNA duplex sequencing. SD, standard deviation; DCS, duplex sequence consensus.

	Young (20 to 29 years)	Intermediate (30 to 34 years)	Old (35 to 42 years)
Number of donors	8	6	8
Average age in years (SD)	25.6 (2.8)	32.7 (1.4)	38.1 (2.7)
Oocytes (DCS depth < 50×)*	28 (0)	25 (2)	27 (0)
Blood samples (DCS depth < 50×)*	6 (2)	5 (2)	7 (3)
Saliva samples (DCS depth < 50×)*	5 (1)	1 (0)	2 (0)

*The number of samples with DCS < 50× is shown in parentheses.

by all DCSs in all tissues studied for an individual), with 8 to 40 such sites per individual (fig. S2). In addition, we called variants—sites with nucleotides different from those present in most DCSs for a sample (see Materials and Methods).

Across all samples studied, we identified 4643 variants. Among them, we detected and analyzed separately (see below) 148 inheritable heteroplasmies [i.e., variants present in all somatic tissues assessed and ≥1 oocyte(s) of a donor]. From the remaining variants, we filtered out (i) 146 variants with ambiguous timing of occurrence during development (i.e., representing potential inheritable heteroplasmies, early somatic mutations, or early germline mutations; see Materials and Methods); (ii) 539 variants present at the beginning or the end of long homopolymer runs (representing potential read mapping errors); (iii) 222 linked variants (with >3 variants present in single paired-end reads mostly occurring in multiple unrelated samples, representing potential regions of nuclear DNA that are homologous to mtDNA, or Numts); and (iv) 63 low-confidence variants present in oocytes with a mean DCS depth of <50× (this way, we excluded variants from two oocytes but were still able to retain variants from one to five oocytes per woman). As a result, we obtained 3525 high-confidence tissue-specific de novo mutations (fig. S3).

To study the frequencies and patterns of de novo mutations and inheritable heteroplasmies depending on age, we assigned the studied women to three age groups of approximately equal size (based on the number of women and the number of oocytes): (i) young (20 to 29 years; eight women; 28 single oocytes), (ii) intermediate (30 to 34 years; six women; 25 single oocytes), and (iii) old (35 to 42 years; eight women; 27 single oocytes; Fig. 1A and table S1). The distribution of 3525 high-confidence de novo mutations among the age groups and tissues is shown in table S3. Most of these de novo mutations (3051) were each measured in a single DCS, and thus had low minor allele frequencies (MAFs). A total of 247 de novo mutations (69, 108, and 70 in oocytes, blood, and saliva, respectively) had a MAF of ≥1%, a common cutoff for mtDNA mutation detection in other studies [e.g., (*17*, *18*)]. A total of 184 of them were supported by single DCSs, and for these mutations, high MAFs likely resulted from low sample sequencing depth. To alleviate this potential limitation, we also identified de novo mutations (and later, also inheritable heteroplasmies), which, in addition to representing high-confidence mutations, also had high-confidence MAFs as supported by ≥3 DCSs (allowing a more reliable MAF estimation). A total of 64 of 247 (59, 3, and 2 in oocytes, blood, and saliva, respectively) de novo mutations satisfied this criterion (fig. S4). Because these mutations are tissue-specific, they likely rose to high frequencies due to clonal expansions. However, we cannot exclude the possibility of multiple independent mutations originating at the same position in the same sample.

De novo mutations

Increase in mutation frequency with age in somatic tissues but not in oocytes

Among all age groups and regions analyzed, oocytes displayed an ~17- to 24-fold lower mutation frequency than blood and saliva (Fig. 1, B to D). To study the effect of age on mutation frequency, we built a generalized mixed-effects linear model considering each individual and each tissue separately (each oocyte was also considered separately; see Materials and Methods), following our approach in (*27*). The model predicted the probability of having a mutation at a site as a function of age (Fig. 1B) and included the mutation frequency as a response, the number of sequenced nucleotides per sample (i.e., sequencing depth per site) as weight, and age, tissue, and their interactions as fixed-effect predictors. The donor ID was used as a random effect. The model indicated a significant increase in mutation frequency with age in blood, a marginally significant increase in saliva, and no difference in oocytes (Z test $P = 2.7 \times 10^{-3}$, $P = 5.7 \times 10^{-2}$, and $P = 0.688$ for blood, saliva, and oocytes, respectively; Fig. 1B and table S4). The donor ID had a marginally significant effect (likelihood ratio test $P = 0.063$), indicating potential differences in mtDNA mutability among different women. The country where a woman was born was not a significant predictor (likelihood ratio test $P = 1.000$) and thus was not added to the final model.

We then compared the mutation frequencies among women from different age groups (i.e., young, intermediate, and old). In agreement with results from the generalized mixed-effects linear model, in blood, the mutation frequency was significantly higher in old compared to young women (i.e., between the two extreme age groups; one-sided permutation test for equality in medians $P = 2.96 \times 10^{-2}$), with a 1.34-fold increase. In saliva, the mutation frequency was marginally significantly higher in old compared to young women (one-sided permutation test for equality in medians $P = 6.66 \times 10^{-2}$), with a 1.32-fold increase. In oocytes, no significant difference in mutation frequency was observed (one-sided permutation test for equality in medians $P = 0.550$), and the mutation frequencies for old versus young women were very similar (0.99-fold decrease; Fig. 1C and table S3). No significant differences were observed in any two adjacent age groups for any tissue. Similar results were obtained when aggregating mutations across women in each tissue and age group (table S5). Such aggregated mutations were used in subsequent analyses of de novo mutations to increase statistical power.

Mutation types and strand bias

Among all tissues and age groups, most de novo mutations were transitions, with transition/transversion ratios of 18.3, 10.3, and 1.7 in blood, saliva, and oocytes, respectively (table S6). The only significant accumulation of a specific mutation type with aging was

observed for A>G/T>C transitions in blood [a previously identified mtDNA-specific mutational signature of aging (*30*)], with a 1.6-fold increase in old versus young women (Fisher's exact test $P = 4.26 \times 10^{-4}$; table S7). The C>T/G>A mutation frequency was 1.6- to 2.8-fold higher at CpG sites than at non-CpG sites in blood and saliva (significant for both tissues and all three age groups except for saliva of old women) but was not significantly different between CpG and non-CpG sites in oocytes (see table S8 for *P* values).

Consistent with previous reports (*19*, *27*, *31*), several mtDNA mutation types in our data exhibited strand bias, defined as a bias in the occurrence of mutations between the light (L) and heavy (H) mtDNA strands. With the L-strand as a reference, we observed a bias of G>A over C>T (4.6- to 39.6-fold), T>C over A>G (2.0- to 3.2-fold), and C>G over G>C (2.3- to 14.3-fold) mutations (fig. S5, A and B, and table S9). When analyzing the D-loop separately, strand bias was evident for transitions in oocytes but not in somatic tissues (fig. S5, C and D).

Mutation frequency and germline mutation rate along the mtDNA

In oocytes, the mutation frequency was significantly higher (6.9- to 8.2-fold) in the D-loop than in the "coding region" (i.e., protein-coding, rRNA, and tRNA regions combined) for each age group studied (Fig. 1D and fig. S6; see table S5 for Fisher's exact test *P* values). Using these data and considering only de novo mutations with MAF ≥ 1% ($n = 52$; only oocytes sequenced at a mean DCS depth of ≥500× were included ensuring that mutations can be measured in ≥3 DCS) and 70 mother-to-oocyte transmissions, we computed the germline mutation rate to be 3.95×10^{-4} [95% confidence interval (CI): 2.68×10^{-4} to 5.60×10^{-4}] and 1.95×10^{-5} (95% CI: 1.21×10^{-5} to 2.99×10^{-5}) mutations per site per generation for the D-loop and coding region, respectively. Thus, the germline mutation rate was 20.2-fold higher in the D-loop than in the coding region. These results are consistent with elevated mutation rates in the D-loop (*16*). In somatic tissues, a significant difference (a 2.7-fold increase) in mutation frequency between the D-loop and coding region was observed only in saliva of young women (fig. S7 and table S5).

The same trend was confirmed when analyzing observed versus expected numbers of de novo mutations based on the length of each mtDNA functional region (table S10). In oocytes, we found the D-loop to harbor a significantly higher number of mutations than expected by chance in each age group analyzed (two-sided binomial test $P = 3.16 \times 10^{-96}$, $P = 1.83 \times 10^{-98}$, and $P = 1.02 \times 10^{-97}$ for young, intermediate, and old, respectively) and the protein-coding region to harbor a significantly lower number of mutations (two-sided binomial test $P = 1.20 \times 10^{-29}$, $P = 6.63 \times 10^{-32}$, and $P = 1.66 \times 10^{-38}$ for young, intermediate, and old, respectively; table S10). Similar trends were observed for somatic tissues but were significant only for a few combinations of age groups and functional regions (table S10).

For de novo mutations located at protein-coding regions, we measured the nonsynonymous-to-synonymous rate ratios, i.e., hN/hS (*32*). In oocytes, such ratios were within the range of neutral expectations (0.90, 1.19, and 0.99 for young, intermediate, and old women, respectively; table S11). In line with this, conservation (measured by phastCons scores) was not significantly different for mutated sites relative to sites without a mutation in the coding region (table S12). In somatic tissues, the hN/hS ratios were higher than expected under neutrality in young women (1.64 in blood and 2.01 in saliva) and in saliva of women from the intermediate age group (2.54), suggesting positive selection; these ratios were not significantly different from neutral expectations in old women (table S11). Positive selection in somatic tissues was also suggested based on conservation using phastCons scores, with odds ratios < 1 indicating that more mutations were observed at conserved sites than at nonconserved sites (table S12). A separate analysis of the D-loop and coding region suggested that this result was mainly driven by the coding region (table S12).

Disease-associated variants

We next assessed whether any de novo mutations were located at sites previously associated with diseases. A total of 109 confirmed disease-associated positions were considered as listed in MITOMAP (*33*) (see Materials and Methods for details). We identified a total of 32 disease-associated variants (0.91% of all mutations) located at 29 different positions (with disease-associated positions representing only 0.66% of the mtDNA; fig. S8). Overall, the proportion of positions affected by mutations was higher for disease-associated versus non–disease-associated sites (26.6%, or 29 of 109, versus 16.3%, or 2685 of 16,460; Fisher's exact test $P = 2.61 \times 10^{-2}$), suggesting high mutation rate and/or positive/destructive selection acting on the former sites. However, lower proportions of disease-associated mutations among all mutations were measured in oocytes than in blood or saliva (0.75, 1.00, and 1.24% of all mutations, or 14, 10, and 8 disease-associated mutations in oocytes, blood, and saliva, respectively), suggesting selection against such mutations in oocytes. Two disease-associated mutations found in oocytes were measured at MAFs ≥ 1% and in ≥3 DCS: m.3946G>A associated with MELAS syndrome (myopathy, encephalopathy, lactic acidosis, and stroke-like episodes) (*34*) and m.1555A>G associated with DEAF (autism spectrum intellectual disability) (*35*). Three additional mutations in somatic tissues had MAFs ≥ 1%. However, they were only measured in a single DCS each, rendering their MAF estimation unreliable.

Variant hotspots

To identify variant hotspots, we built a probabilistic model that considers tissue-specific estimates of the average mutation frequencies, the high mutability in the D-loop, and the mean DCS depth in each sample (*27*). On the basis of the model results (Fig. 2A), we defined variant hotspots as sites mutated in somatic tissues of at least three women or in oocytes of at least four women. We detected a total of 259 de novo mutations at 73 hotspots: 51 mutations at 22 hotspots in blood, 19 mutations at 15 hotspots in saliva, and 189 mutations at 36 hotspots in oocytes (the number of tissue-specific hotspots depends on the total number of mutations measured in each tissue and thus is higher in oocytes). Fifteen hotspot sites overlapped between two different tissues (but never among all three), resulting in a total of 58 affected mtDNA sites (table S13). Overall, as many as 3, 5, and 10% of all de novo variants in saliva, blood, and oocytes, respectively, were found at hotspot sites occupying only ~0.4% of the mtDNA length (58/16,569).

Among the 58 hotspot sites, 40 were located in the D-loop, 4 in rRNA, 4 in tRNA, and 10 in protein-coding regions—a distribution that is different from what is expected based on the length of mtDNA regions as well as on the distribution of mutations occurring at non-hotspot sites (Fig. 2B). In particular, hotspots were significantly overrepresented in the D-loop, consistent with its high mutation rate (*16*) and significantly underrepresented in the protein-coding regions (table S10D). In oocytes, 35 of the 36 hotspots were located in the D-loop. In contrast, in somatic tissues, only 5 of the 22 hotspots were located in the D-loop, but 10 were in protein-coding regions (all

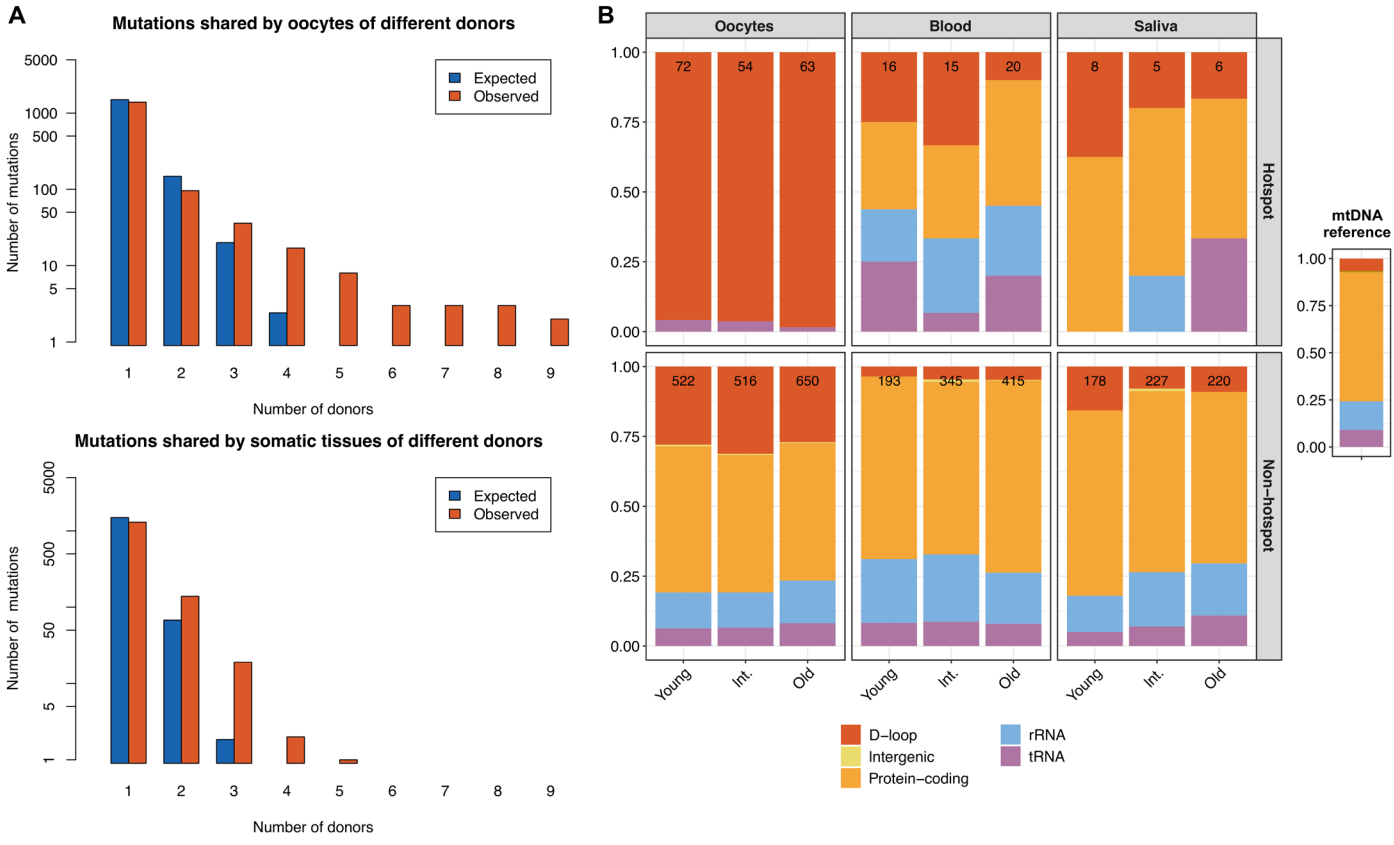


Fig. 2. Analysis of variant hotspots. (**A**) Observed and expected distributions of the number of women sharing individual variants in oocytes (with all oocytes from a donor considered together) and somatic tissues. Because of similar mutation frequencies and lower numbers of measured mutations, blood and saliva were considered together as somatic tissues. We also considered the number of sequenced oocytes per donor to evaluate oocyte variant hotspots. For oocytes and somatic tissues separately, we computed the expected number of mutations present in exactly one woman and shared by several women. Our results suggest that, due to random chance, mutations at the same site can occur in two women but are rarely expected to occur in somatic tissues of three or more women and in oocytes of four or more women. (**B**) Distribution of mutations (normalized by the length of the respective region) among D-loop, intergenic regions (i.e., noncoding DNA outside of the D-loop), protein-coding regions, rRNA, and tRNA, shown separately for each tissue and hotspots versus non-hotspots. Numbers indicate the total number of mutations analyzed. The proportion of mtDNA occupied by each region is shown on the right.

nonsynonymous), 4 in rRNA, and 3 in tRNA. These results corroborate our findings of high de novo mutation frequency in the D-loop, independent of age, specifically for oocytes (Fig. 1D and table S5). Only one variant hotspot (somatic) overlapped with a disease-associated mutation: m.3243A>G, a mutation in tRNA associated with MELAS syndrome, one of the most mutable positions in the mtDNA genome (*16*) suggested to be subject to positive selection (*36*).

Inheritable heteroplasmies

In addition to de novo mutations, we investigated heteroplasmies present in somatic tissues and at least one oocyte of the same woman (table S14) and thus were likely already inherited from her mother. Because of their usually higher MAFs, such heteroplasmies have a higher potential to be inherited by the offspring; we called them "inheritable heteroplasmies." To boost our statistical power and confidence in correctly classifying them, in addition to the samples used to study de novo mutations, here we also considered samples with average DCS depth below 50×. We also used SSCSs and additional targeted duplex sequencing (see Materials and Methods) for blood and saliva to aid us in distinguishing between early germline mutations and inheritable heteroplasmies. MAFs measured using exonuclease V digestion and targeted mtDNA capture enrichment were highly correlated (Pearson $r = 0.995$, $P = 1.61 \times 10^{-12}$; fig. S9).

In our dataset, 19 women harbored 63 sites (one to nine sites per woman; fig. S10) with inheritable heteroplasmies (13, 22, and 28 sites in young, intermediate, and old women, respectively; tables S1 and S14), which were all transitions. A total of 45 sites were located in the D-loop, 13 in protein-coding regions, and 5 in rRNA. A total of 14 sites overlapped variant hotspots in oocytes and 3 overlapped variant hotspots in somatic tissues; all of them were located in the D-loop. Only one inheritable heteroplasmy was associated with a disease [m.10197G>A associated with Leigh disease, dystonia, stroke, and Leber optic atrophy and dystonia (*33*)].

MAFs of inheritable heteroplasmies were correlated between oocytes and somatic tissues (Pearson $r = 0.802$, $P = 2.66 \times 10^{-15}$; fig. S11A), as well as between blood and saliva (Pearson $r = 0.802$, $P = 6.30 \times 10^{-5}$; fig. S11B). Similar results were obtained when including only heteroplasmies with high-confidence MAFs (i.e., those measured in at least three mtDNA copies; 30 heteroplasmic sites; fig. S12, A and B).

Heteroplasmy variance does not depend on age but depends on MAF

We next tested whether age had an effect on heteroplasmy variance, i.e., whether the normalized variance in MAF [variance divided by $H(1 - H)$, where H is the average MAF of the allele among single oocytes or across somatic tissues (*37*, *38*)] increases with age. We did not observe a significant correlation between the normalized variance in MAF within each sample type and age ($P = 0.256$ and $P = 0.907$ for somatic tissues and oocytes, respectively; fig. S11, C and D). Results were similar when including only heteroplasmies with high-confidence MAFs (fig. S12, C and D).

To estimate the size of the effective germline bottleneck—the size required to explain observed heteroplasmy variance—for human mtDNA, we applied the population genetics approach described in (*38*, *39*) to the data on MAF shifts at 63 inheritable heteroplasmies measured in 19 women. Note that this approach does not take into account nonrandom distributions of variants and heteroplasmy levels due to selective forces, which should be modeled in future studies. Namely, we compared allele frequencies between somatic tissues (averaged among the tissues sequenced per donor for donors with more than one tissue sequenced) and single oocytes (with two to five oocytes per donor; donor hs001 with one oocyte did not have an inheritable heteroplasmy) for each site separately (245 transmissions to oocytes in total; table S15). The effective bottleneck size was estimated to be 909.39 segregating mtDNA units (95% bootstrap CI: 0.47 to 5329.73). The effective germline bottleneck computed based on heteroplasmies with high-confidence MAFs (30 heteroplasmies and 111 transmissions) was similar in magnitude: 1040.03 segregating mtDNA units (95% bootstrap CI: 0.34 to 11,089.05). Large CIs reflect the large variation in bottleneck size calculated for heteroplasmies measured at low MAFs.

To evaluate whether heteroplasmy MAF affects our estimates of the bottleneck size, we also separately analyzed high-frequency heteroplasmies, i.e., the ones in which at least one variant had MAF $\geq 1\%$. In total, 27 of the 63 heteroplasmic sites (belonging to 17 women) satisfied this criterion. The effective bottleneck size estimated for such sites based on 100 transmissions to oocytes was 30.41 segregating mtDNA units (95% bootstrap CI: 0.02 to 178.53)—which was in line with the recent estimates [e.g., summarized in (*16*)]. Thus, the effective bottleneck size estimates depend on the MAF of heteroplasmic sites. Considering each age group separately, we estimated the effective bottleneck size to be 25.03 units (95% bootstrap CI: 0.47 to 98.42) for young women, 38.60 units (95% bootstrap CI: 0.02 to 74.42) for women in the intermediate age group, and 28.38 units (95% bootstrap CI: 0.38 to 178.53) for old women (table S15). Thus, the bottleneck size was not significantly different among women in different age groups.

Selection affects allele frequencies of inheritable heteroplasmies

Next, we analyzed whether selection influences changes in MAFs for inheritable heteroplasmies. We referred to the minor allele recorded in the somatic tissues as the minor allele of a heteroplasmic site in a donor and calculated the MAF difference between oocytes and somatic tissues. A positive (negative) change represents a higher (lower) MAF in oocytes compared to somatic tissues. We observed a significantly lower number of heteroplasmic variants with a positive ($n = 56$) versus negative ($n = 189$) change ($P = 4.75 \times 10^{-18}$, binomial test; Fig. 3, A and B). The same remained true when considering only heteroplasmies in protein-coding regions (9 with a positive change and 41 with a negative change, $P = 2.16 \times 10^{-5}$), in the D-loop (44 with a positive change and 130 with a negative change, $P = 6.53 \times 10^{-10}$), and in rRNA (3 with a positive change and 18 with a negative change, $P = 3.48 \times 10^{-3}$). Among heteroplasmies in protein-coding regions, a significantly higher number of negative shifts was observed for nonsynonymous ($P = 2.16 \times 10^{-5}$) but not for synonymous variants ($P = 0.418$; Fig. 3B). The same was true when including only heteroplasmies with high-confidence MAFs (fig. S13). These results suggest purifying selection against heteroplasmies transmitted to oocytes and/or positive selection in somatic tissues.

Heteroplasmic shifts calculated as $\log_2$ fold change in MAFs (*40*) reached levels around ± 10. This represents an ~1000-fold change in MAF from a woman to her oocyte (Fig. 3C), consistent with an ~1000-fold reduction in mtDNA content during human germ cell development (*41*). However, most heteroplasmies had a much smaller change in MAF from a woman to her oocyte, with a <64-fold change in 94.1% of the heteroplasmic variants. Because we used the minor allele recorded in the somatic tissues as the minor allele of a heteroplasmic site in a donor, our results suggest again that purifying selection acts on inheritable heteroplasmies in oocytes or, alternatively, positive selection acts in somatic tissues (or a combination of both).

Frequency-dependent selection on mtDNA variants in oocytes

We next analyzed the distribution of oocyte genetic variants across the mtDNA molecule depending on their frequency and inferred heritability by considering three nonoverlapping categories (Fig. 4 and table S16): (1) de novo mutations with MAF < 1%; (2) de novo mutations with MAF $\geq 1\%$ (mean MAF = 3.07%, SD = 3.96%); and (3) inheritable heteroplasmies (mean MAF = 2.07%, SD = 8.90%). Although, for all categories, lower proportions of mutations were found in the coding region than expected based on the size of this region, we found that most variants in the first category were located in the coding region (protein-coding, rRNA, and tRNA regions); variants in the second category were distributed approximately equally between the D-loop and the coding region (with slightly more mutations in the D-loop); and variants in the third category were predominantly located in the D-loop. The distribution of mutations inside versus outside the D-loop was significantly different between de novo mutations with MAF < 1% and those with MAF $\geq 1\%$ ($P = 2.04 \times 10^{-3}$, chi-square test), as well as between de novo mutations with MAF < 1% and inheritable heteroplasmies ($P = 1.75 \times 10^{-8}$, chi-square test), but not between de novo mutations with MAF $\geq 1\%$ and inheritable heteroplasmies ($P = 0.111$, chi-square test). This pattern was shared by the three age groups considered and suggests selection operating against coding mutations in oocytes, especially for variants with high frequency in oocytes and for inheritable heteroplasmies. In line with this, conservation analysis [measured by phastCons scores (*42*)] also suggests higher selective pressure on variants with high MAFs (table S17). Whereas, in all three MAF categories, more mutations were observed at nonconserved sites than at conserved sites, the difference was more pronounced in the second and third category, reflected by higher odds ratios (odds of observing a mutation at nonconserved site/odds of observing a mutation at a conserved site), especially in the intermediate and old age groups. When separately analyzing conservation in the D-loop and coding region, we did not observe any significant indications of selection in categories (2) and (3) but were limited by small numbers of mutations in the individual regions (table S18). In category (1), analysis suggests purifying selection in the D-loop but becomes nonsignificant

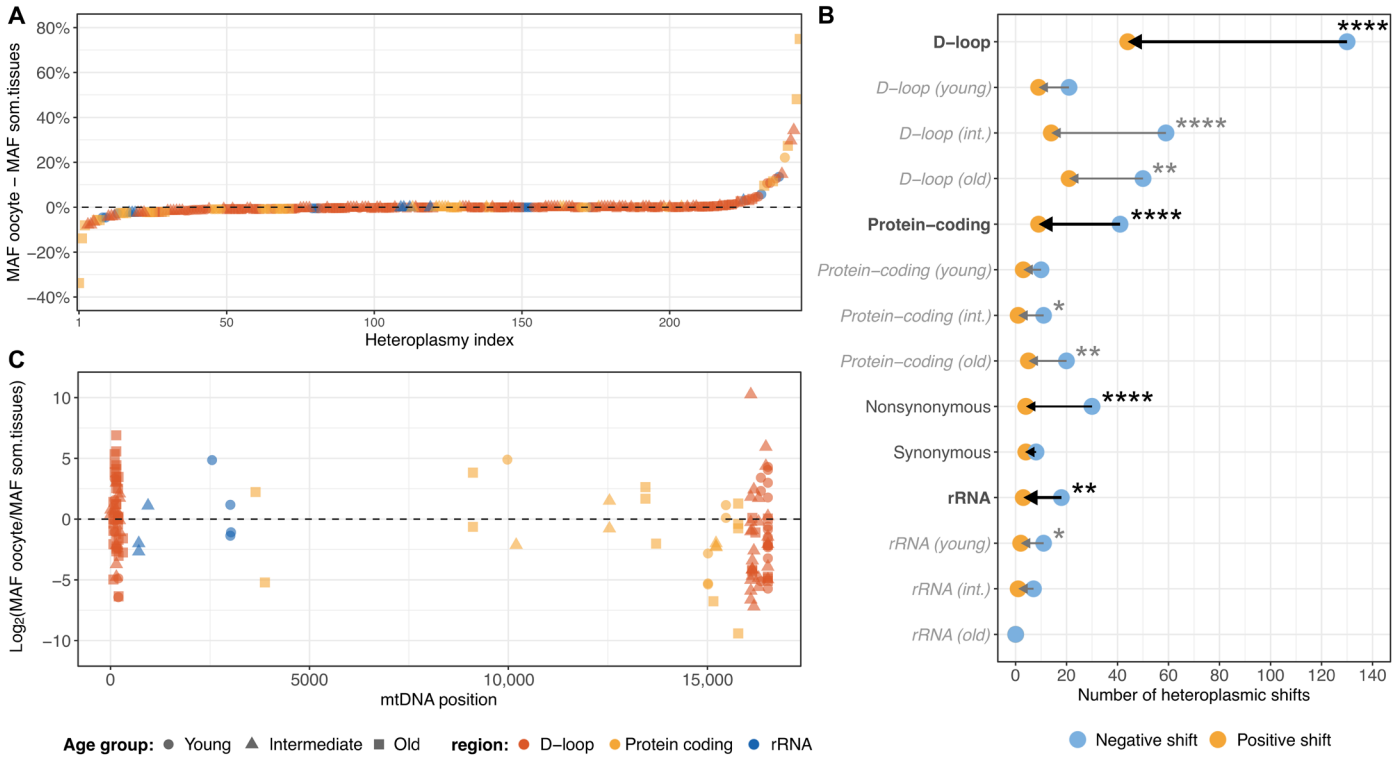


Fig. 3. Evaluation of selection in heteroplasmic transmissions. (**A**) MAF difference between oocytes and somatic tissues measured in percentage points; a positive (negative) shift represents a higher (lower) MAF of a heteroplasmy in oocytes compared to somatic tissues for a given a woman. (**B**) Number of heteroplasmies showing an increased or decreased MAF in oocytes (positive or negative shift, respectively), shown separately for mtDNA regions, and either combined or separated by age group. Left-pointing arrows indicate higher numbers of negative versus positive shifts. *$P < 0.05$, **$P < 0.01$, and ****$P < 0.0001$; binomial test, corrected for multiple testing using the method of Benjamini-Hochberg to control the false discovery rate. (**C**) Log_2 ratio of MAF difference between oocytes and somatic tissues, calculated to take into account the magnitude of the fold change in MAFs (*40*). A total of 109 transmissions with MAF = 0 in an oocyte are not displayed due to the resulting negative infinite values. Colors indicate their localization in different mtDNA regions, and shapes show the age group of a woman.

after downsampling to similar numbers as assessed in the other categories; higher numbers of mutations would be required for proper comparisons between frequency groups. However, mitochondrial local constraint (MLC) scores (*43*), a recently developed annotation metric based on local constraint to assess the local tolerance to base or amino acid substitutions, were higher in category (1) (average = 0.39, SD = 0.30), compared to categories (2) and (3) (average = 0.21 and 0.18, SD = 0.24 and 0.23, respectively; table S19). The MLC score assigns each variant a score ranging from 0 to 1, with 1 being the most constrained variant and therefore likely to have the most deleterious impact when mutated. Therefore, our results suggest lower selective pressure against deleterious mutations when present at lower frequencies.

Variants with known disease association always showed higher average MLC scores than other variants and had MLC scores ≥ 0.5, a suggested cutoff for deleteriousness (*44*) (table S19).

DISCUSSION

In our study of mtDNA mutations, we used both germline (single oocytes) and somatic tissues from the same women. This provided us with several unique opportunities. Directly measuring mutations in single oocytes allowed us to disentangle mutations and selection in oocytes from post–fertilization selection. Also, sequencing at least one somatic tissue in addition to single oocytes allowed us to discriminate between de novo somatic and germline mutations, as well as between de novo germline mutations and mutations already segregating in the germ line.

In addition, using duplex sequencing, which has an inherently low error rate ($<10^{-7}$) (*19*), allowed us to comprehensively study mtDNA variants with various allele frequencies, including those with MAF < 1%, the latter constitute 96.9% of all de novo mutations in oocytes in our dataset. This provided a detailed picture of the mtDNA mutational frequency and landscape in oocytes. Because of methodological limitations, previous studies [e.g., summarized in (*16*)] considered only mtDNA variants with frequencies ≥ 1% (or even higher). This rather high MAF cutoff is particularly problematic for oocytes. Because of the very high copy number in human oocytes (~100,000 on average) (*45*), even for a variant observed at a MAF of 1%, ~1000 mtDNA molecules would be affected in a single cell. Although the high mtDNA copy number might have a compensatory effect, as suggested previously (*46*), very-low-frequency variants can have considerable impact on mitochondrial function and be substantially affected by selection—yet they were excluded from most previous studies.

Age does not increase mtDNA mutations in human oocytes

Analyzing women ranging in age from 20 to 42 years, both based on individual age and on age groups, we did not observe an increase in mutation frequency in the oocytes in older women. This finding is

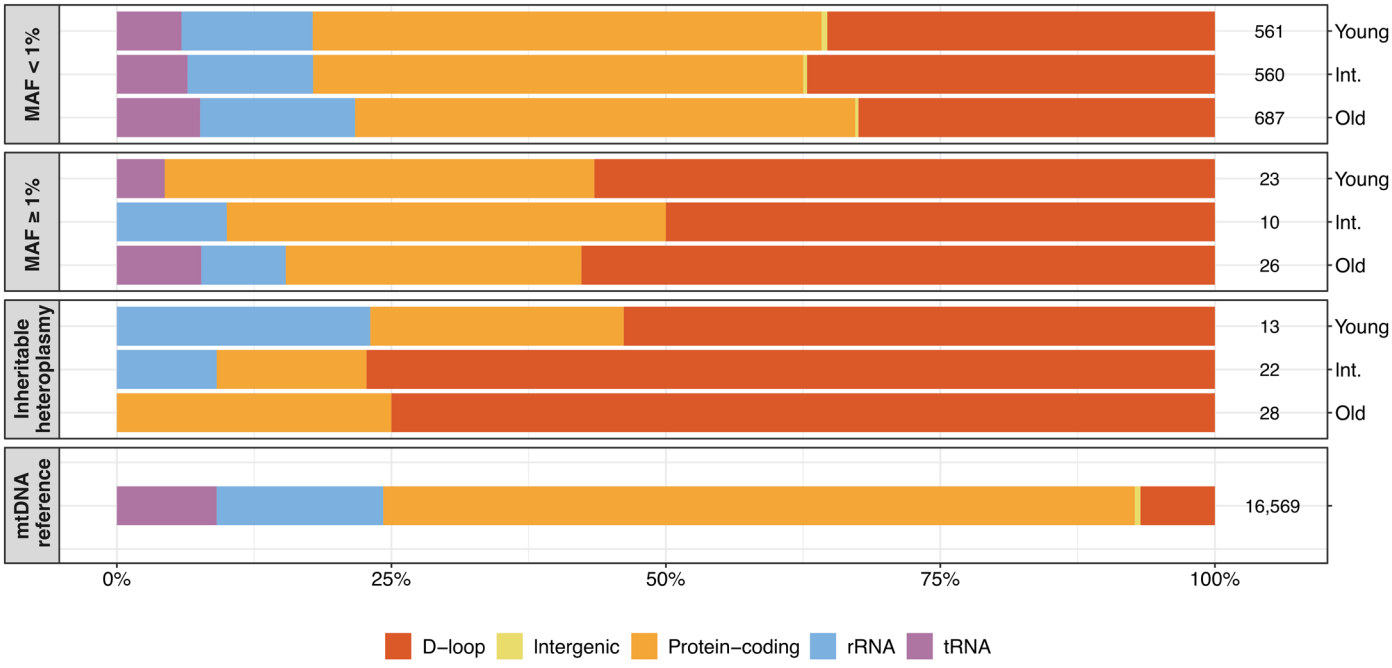


Fig. 4. Proportions of mtDNA variants in oocytes depending on their frequency, heritability, and age group. Distribution of mutations (normalized by the length of the respective region) among D-loop, intergenic DNA, protein-coding regions, rRNA, and tRNA, shown separately for de novo mutations with MAF < 1%, de novo mutations with high-confidence MAF ≥ 1% (measured in ≥3 DCS to ensure proper representation on MAFs; 10 mutations measured in <3 DCS were excluded), and inheritable heteroplasmies. Numbers on the right represent the total number of mutations analyzed or the number of base pairs annotated in each region for the mtDNA reference.

unlikely to be an artifact from limited numbers of mutations assayed because we (i) detected similar numbers of mutations for oocytes and for somatic tissues (table S3) and (ii) sequenced at least an order of magnitude more nucleotides for oocytes than for somatic tissues (table S5) yet could detect a significant increase in mutation frequency only in somatic tissues (see below) but not in oocytes.

The lack of mutation frequency increase with age in humans differs from what we observed using the same methods (i.e., duplex sequencing) in mouse oocytes, where we found ~2-fold higher mutation frequencies in 10-month-old compared to ~1-month-old mice (*26*). However, for rhesus macaques, again using the same methods, we found that mutation frequencies in oocytes increased only until the age of ~9 years and not further (*27*) [as rhesus macaques age ~3 times faster than humans, ~9 years equivalent to ~27 years for humans (*47*)]. This suggests that oocytes of older macaques maintain the quality of their mtDNA, which might be also true for humans. Because our samples were obtained from women undergoing in vitro fertilization (IVF) treatments, the age of donors was limited to 20 to 42—a subset of the human female reproductive life span [~13 to ~50 years (*48*, *49*), with fertility peaking around age 24 and diminishing after 30 (*50*)]. We therefore cannot exclude the possibility of an increase in mtDNA mutations at an earlier age in humans, similar to what we observed in macaques, sampled by us starting from the age of 1 year (equivalent to ~7 years in humans). Further studies including younger women will be required to study such a potential early age effect. Furthermore, increased mtDNA mutations may occur at more advanced age; however, mitochondria with a high number of deleterious mutations might be removed by mitophagy [reviewed in (*51*)], or oocytes with high mtDNA mutational load might be eliminated by follicular atresia [reviewed in (*52*)].

Results from previous studies directly assessing mtDNA mutation increases in human oocytes have been contradictory and might have been affected by PCR and sequencing errors as no previous studies used highly accurate duplex sequencing. Nevertheless, our findings are in agreement with a study analyzing 53 human oocytes, which identified 17 variants with MAF > 2% and showed no increase in their frequency with ovarian aging (*23*). In contrast, Yang and colleagues reported higher numbers of mtDNA mutations with MAF > 0.2% in oocytes from older versus younger women (*53*). Their study analyzed oocyte pools (and not single oocytes) with an unreported number of oocytes in each pool; a higher number of oocytes in pools of older women could have led to a higher number of mutations. In general, using a less accurate method, they observed >8000, or ~4 times more mutations in 76 oocytes as compared to our observations of 1877 mutations in 78 oocytes, suggesting the presence of sequencing errors in their study.

Another study (*24*) showed a correlation between age and the number of mtDNA de novo mutations with MAF ≥ 2%. A possible explanation for this discrepancy in results might be a different selective pressure acting on variants measured at higher MAFs and/or a change in this pressure with age. To evaluate this for our data, we also considered only mutations with MAF ≥ 2%; these mutations do appear to increase with age but nonsignificantly. When including only 22 oocyte mutations with MAF ≥ 2% (to ensure an accurate representation of MAFs, only the 74 oocytes with a mean DCS depth of >200× were included in this analysis), we found 7, 5, and 10 mutations analyzing 24, 23, and 27 oocytes, in the young, intermediate, and old age groups, respectively. This represents a 1.27-fold higher, not significant (Fisher's exact test $P = 0.809$) mutation frequency increase in old versus young women. Larger numbers of mutations would be required to gain statistical power.

Somatic mutations increase with age

Although we had limited power to study somatic mutations in our data because of low mtDNA enrichment and sequencing depth (partly explained by the low mtDNA copy number in these samples), we found a significant and marginally significant increase in mutation frequency in blood and saliva, respectively. Our observed mutation frequencies are in the same range as previous reports in human blood (*54*) and other somatic tissues (*19*, *31*). Neither in blood nor in saliva, we observed a strong enrichment of mutations in the D-loop, which is in contrast to germline mutations as well as mutations measured in the brain (*19*) but is in agreement with previous reports for human lymphocytes and monocytes (*55*), suggesting tissue-specific differences in mtDNA mutational patterns (*56*).

Mechanisms of mitochondrial mutations

The predominance of transition mutations, combined with strand bias, in all tissues assessed is consistent with a previously suggested major contribution of replication-associated errors in generating mtDNA mutations (*16*, *19*, *57*, *58*). Prior studies have shown that the polymerase responsible for mtDNA replication (DNA polymerase gamma) has a propensity for transition mutations (*59*–*61*). Another potentially contributing mechanism is the spontaneous deamination of cytosine (C>T) and adenosine (A>G) (*62*, *63*)—also leading to transitions. The strong strand bias we observed for G>A over C>T and for T>C over A>G on the reference L-strand—especially outside of the D-loop—is consistent with a high incidence of C>T and A>G mutations occurring on the H-strand. A similar strand bias was reported in other studies (*16*, *19*, *57*, *58*). This high incidence of mutations on the H-strand might be explained by its single-stranded status during the initial stages of replication [reviewed in (*64*)], which facilitates DNA damage, potentially accounting for the nucleotide content differences between the H- and L-strands (the H-strand is guanine- and thymine-rich). Note that, in oocytes, the presence of strand bias in the D-loop is inconsistent with a replication-driven mutation pattern expected based on the bidirectional mtDNA replication operating there (*57*, *65*). As somatic cells from older women have undergone more cycles of replication and hence spent more time in a single-stranded state, our observation of A>G/T>C transitions being the only mutation type with a significant age-associated increase in blood, coupled with increased strand bias in older women, indicates a contribution of replication to their age-associated mutagenesis. Furthermore, our observation of higher rates of C>T/G>A transitions at CpG than non-CpG sites in somatic tissues suggests a role of spontaneous deamination of methylated cytosines (*62*) in mtDNA mutagenesis, despite controversial reports regarding CpG methylation in mtDNA [e.g., (*66*–*70*)]. This pattern was not observed in oocytes.

Does selection affect the germline and somatic mtDNA mutational landscape?

We did not find strong evidence of selection operating on low-frequency mtDNA de novo mutations in protein-coding regions of oocytes. Such mutations might be subject to less selective pressure due to their low frequency, at which they are not expected to have phenotypic effects on a cellular or mitochondrial level. Consistent with this, the distribution of low-frequency (MAF < 1%) mutations in oocytes between coding and noncoding regions of mtDNA was more in line with expectations based on the lengths of these regions than that for mutations with higher frequency (Fig. 4 and table S16). Moreover, such mutations were more frequently found at conserved sites (table S17) and at more deleterious sites (table S19). Nevertheless, we observed a lower incidence of disease-associated mutations in oocytes compared to somatic tissues. Thus, purifying selection might be acting against such mutations in oocytes. However, the selective pressure on individual mutations might be different in specific somatic tissues or cell types compared to oocytes, and low-frequency mutations in oocytes might rise to high frequencies during development.

When we analyzed germline mutations with higher frequencies, we observed a shift in their distribution from coding toward noncoding regions. This was the case for de novo mutations with MAF ≥ 1 and for inheritable heteroplasmies, suggesting frequency-dependent purifying selection in the germ line. Specifically, selection would act against higher frequency functional variants, i.e., variants located in coding regions. Consistent with this, previous reports of selection have been based on mtDNA variants with higher frequencies (≥1 to 5%, observed in pedigrees) (*16*–*18*, *40*), mainly reporting purifying selection in the female germ line. Differences in selective pressure depending on variant MAF were also reported in mice (*46*).

Age-dependent selection after fertilization acting on variants occurring at higher MAFs could explain why some other studies have reported increased numbers of mutations in children of older mothers (*17*, *18*). Selective forces acting after fertilization [such as replication advantages of specific variants (*71*)] and changes in selection pressure with aging, leading to increased numbers of variants measured at higher MAFs, might be responsible for this observation instead of increased oocyte mutagenesis per se. Consistent with this hypothesis, López-Otín and colleagues have previously suggested that the mtDNA quality control system in cells is less efficient at purging pathogenic mutations (or organelles with such mutations) with advancing age (*72*). Duplex sequencing of pedigree mutations would allow one to include variants at lower frequencies, which may further elucidate the causes of differences observed between our low-MAF oocyte variants and published pedigree data.

In contrast to the observed purifying selection acting on (high-frequency) variants in the germ line, our observation of increased hN/hS ratios in blood and saliva, as well as a high incidence of mutations at conserved sites are indicative of positive selection. This agrees with previous studies, reporting positive selection in human liver (*32*) and macaque liver and skeletal muscle (*27*), and therefore appears to result from a tissue-type independent mechanism. Alternatively, depending on the impact of these mutations on mitochondrial functions, destructive selection (i.e., selection promoting an increase in abundance of deleterious mutations) (*73*) might explain this observation. In agreement with this, we observed high prevalence of disease-associated mutations, suggesting positive or destructive selection acting on these sites, especially in somatic tissues. We do not observe negative selection, as previously suggested in various somatic tissues (*74*).

Furthermore, high-frequency inheritable heteroplasmies were observed in a higher proportion of women as compared to the proportion of oocytes, which is indicative of positive selection in somatic tissues and leads to increases in the MAFs of variants. A total of 54.5% of women in our study carried an inheritable heteroplasmy measured at a MAF of ≥1% (in blood and/or saliva), which is in agreement with previously reported figures for mothers (47.8%) and offspring (42.5%) (*40*). In contrast, only 35.0% of all oocytes we

assessed carried a heteroplasmy above this MAF threshold. This is also in agreement with our observation that, lowering the MAF, one finds higher numbers of inheritable heteroplasmies in oocytes compared to somatic tissues. MAFs measured in somatic tissues might increase with age, leading to more high-frequency variants in older people. In line with this, Liu and colleagues reported higher numbers of high-frequency variants (MAF ≥ 1%) in somatic cells of older versus younger mice (*75*).

The effective germline bottleneck size depends on mutation frequency

By studying inheritable heteroplasmies in oocytes, we were able to estimate the size of the effective germline mtDNA bottleneck directly in human germ cells, resulting in an estimate that is less affected by age-related genetic drift and selection in somatic tissues. Furthermore, the use of highly accurate duplex sequencing allowed us to include in the estimation heteroplasmies with MAFs < 1%. Considering all inheritable heteroplasmies, we estimated the effective germline bottleneck size to be ~909 segregating units (CI: 0.47 to 5329.73). Considering only heteroplasmic sites for which at least one sample had MAF ≥ 1%, we estimated a bottleneck size of ~30 segregating units (CI: 0.02 to 178.53). This is similar to previous estimates in humans ranging between ~3 and 32 (*16*–*18*, *76*) using MAF cutoffs of 1% or higher. The study using the highest MAF cutoff (only including heteroplasmies with MAFs between 10 and 90%) yielded the smallest bottleneck size estimate of around 3 (*16*). The similarity between our bottleneck size estimates based directly on oocytes and prior estimates based on pedigrees suggests that the calculation is only negligibly affected by genetic drift in somatic tissues. Differences in bottleneck size estimates depending on MAF cutoffs suggest differences in selective pressure acting on variants with different frequencies. Considering the organization of multiple mtDNA copies in individual mitochondria and cells, selection likely has a different impact when a larger proportion of mtDNA molecules are affected, which is the case for variants at higher MAFs. Note that wide, overlapping CIs have to be considered in these comparisons.

Germline mtDNA mutation rate estimates

To compare germline mutation rate estimates between our study and prior ones, we computed a germline mutation rate of 3.95×10^{-4} (95% CI: 2.68×10^{-4} to 5.60×10^{-4}) mutations per position per generation for the D-loop, including only de novo mutations measured at MAF ≥ 1%. This is about 5× higher than a previous estimate by Wei and colleagues based on a pedigree analysis using a similar MAF cutoff (*40*). When including only mutations measured at MAFs ≥ 5%, our mutation rate estimate for the coding region is 2.50×10^{-6} (95% CI: 5.16×10^{-7} to 7.32×10^{-6}) mutations per position per generation—which is in the same range as the estimate produced by Árnadóttir and colleagues based on pedigrees and using the same MAF cutoff (*16*) (2.87×10^{-6} mutations per position per generation; 95% CI: 2.79×10^{-6} to 2.95×10^{-6}). Again, to obtain a more complete picture of potential differences in mutation rate estimated based on oocytes and pedigree studies, duplex sequencing of pedigrees could also aid in addressing mutations at lower MAFs, which are currently excluded from such studies.

Potential caveats

When calling de novo mtDNA mutations, we used several experimental and computational steps to minimize the contribution of Numts [e.g., an mtDNA enrichment strategy not prone to coenrichment of Numts, as previously described in detail for macaques (*27*)]. The enzymatic digestion of linear nuclear DNA, the comparatively long fragments after shearing, and the relatively high read length during sequencing, as well as retaining only reads mapping to mtDNA, all minimize the contribution from Numts to our mutation dataset. Moreover, although we cannot completely exclude their contribution, Numts should affect each woman similarly and not bias our comparison of mutations between the three age groups. Considering the efficient enrichment of mtDNA molecules in oocytes (on average, 86.1% of all DCS reads represented mtDNA sequences), we do not expect Numts to affect any interpretations considerably. Furthermore, when inferring MAFs of inheritable heteroplasmies, their frequencies were often based on small numbers of measured variants at a position in a sample. Although, because of the low error rate of duplex sequencing, we can be confident that the variant is present in a sample, MAFs might be over- or underestimated. Moreover, for calculating the effective germline bottleneck and evaluating selection at inheritable heteroplasmies, maternal blood and/or saliva were used as reference tissue(s). Estimates might therefore be biased by selection not only in these tissues but also within oocytes. Note that, unlike for mother-child analyses (*16*), our estimates of the effective germline bottleneck size are not affected by selection in somatic tissues of the offspring, although a bias from selection at the oocyte level (as suggested in our analyses) cannot be excluded. Furthermore, the frequency in the assessed somatic tissue(s) might not represent the average of all somatic tissues [blood was previously shown to consistently have lower heteroplasmy levels compared to skin and urine (*71*)]. Nevertheless, we still observe a significantly lower number of heteroplasmic variants with a positive versus negative change in heteroplasmy frequency (with a negative change representing a lower MAF in oocytes compared to somatic tissues); thus, we might even underestimate the effect of selection. Furthermore, it has to be mentioned that, whereas oocytes represent a single cell type, saliva and blood constitute a mixture of different cell types. This might influence a direct comparison of these tissues. Although such a direct comparison of single somatic cells would be highly beneficial, performing their duplex sequencing (required for a high-confidence mutation assessment) is currently challenging because of low mtDNA copy numbers in most cell types, despite important recent methodological advances [e.g., (*77*, *78*)].

MATERIALS AND METHODS

Experimental design, study participant details, samples, and their handling

All parts of this study were designed and conducted in compliance with the relevant regulations regarding the use of human study participants, and in accordance with the Declaration of Helsinki. Duplex sequencing was performed for somatic tissues and oocytes of 22 women who were scheduled for intracytoplasmic sperm injection (ICSI) at the Kinderwunsch Zentrum (fertility center) of the Kepler University Clinic, Linz, Austria. The women spanned an age range from 20 to 42 years. The study was approved by the Ethics Commission of Upper Austria (F-11-17) and followed the rules of informed consent, anonymization of samples, and avoidance of any ethical sensitivity that may harm an individual. In total, 80 single oocytes were sequenced. Moreover, upon availability and exonuclease V enrichment success, blood from 18 donors and/or saliva from

8 donors (for 4 donors, both samples were available) were sequenced (table S1). For an additional three women (hs001, hs002, and hs004), saliva was sequenced using the targeted mtDNA capture approach; those sequences were only included in the analyses of inheritable heteroplasmies.

As described previously (*26*, *27*), to avoid cross-sample contamination, all pre-PCR steps were performed in designated laminar flow hoods located in a separate room. To minimize the time of sample processing with an open lid, individual samples were handled in separate low-binding tubes (Axygen). Only after DNA amplification, samples were transferred to a 96-well plate format during bead purifications with the epMotion automated liquid handling system (Eppendorf). Afterward, samples were transferred back to separate tubes, minimizing potential contamination.

Sample preparation

Single oocyte collection

The 22 ICSI female patients included in this study either had male partners who suffered from subfertility (59.1%) or themselves suffered from infertility that was not expected to cause germline mutations (e.g., bilateral tubal blockage or anovulation; 40.9%). Controlled ovarian hyperstimulation was performed. For the vast majority of patients ($n = 19$, 86.4%), this was done using a short antagonist protocol, whereas three patients were treated with a long agonist regimen. On average, women were stimulated with 1913 (± 777) IU of purified (Menopur, Ferring; Fostimon, IBSA) or recombinant (Puregon, MSD) follicle-stimulating hormone and stimulation took 11 (± 2) days. Ovulation was induced with 10,000 IU hCG (Pregnyl, MSD). Oocyte retrieval was carried out transvaginally under ultrasound guidance 36 hours after ovulation induction.

Cumulus-oocyte complexes were cultured in GM501 Wash medium (Gynemed) until further processing. Because ICSI requires removal of surrounding cumulus cells from the zona pellucida (ZP) of the oocytes, enzymatic digestion of the same was performed for 30 s (GM501 hyaluronidase, Gynemed, 80 IU). Remaining cumulus cells still obstructing the view were then removed mechanically until the maturity of the gamete could be evaluated properly.

In principle, all metaphase II (MII) oocytes were considered for ICSI and could not be used for study purposes. The only exception was a dysmorphism called "cluster of the smooth endoplasmic reticulum" (*79*). Such affected MII oocytes ($n = 13$) were never injected but instead were transferred into a vial containing 2 µl of phosphate-buffered saline and immediately stored at −20°C.

All immature oocytes were potential research objects. In these cases, culture was attempted overnight to have the immature gametes reach MII stage. All cumulus cells that did not automatically detach through the manipulation process conducted the day before were removed using a diode laser (Fertilase, Vitrolife), so that, finally, the ZP of the gametes was completely free of somatic cells (this was done to exclude potential mutational bias coming from those cells). In cases where telophase I (TI) or MII was reached (as assessed by the presence of the first polar body within the perivitelline space), the ZP was opened by a series of laser pulses so that the first polar body could be biopsied by means of a glass pipette (15 µm in diameter, Gynemed). Polar bodies and oocytes were separately frozen as described above. As a result, oocytes included in this study comprise MII oocytes ($n = 62$), germinal vesicles ($n = 4$), metaphase I oocytes ($n = 8$), and TI oocytes ($n = 6$). All oocytes were initially stored at −20°C and later transferred to −80°C for longer-term storage until further analysis. No oocytes included in this study were ever exposed to sperm.

DNA extraction and mtDNA enrichment from somatic tissues

Total DNA was extracted from 6 to 9 ml of EDTA-blood using the FlexiGene DNA Kit (Qiagen) according to the manufacturer's instructions. Saliva samples were collected using OG-500 tubes (DNA Genotek), and total DNA was extracted using the prepIT L2P Kit (DNA Genotek) according to the manufacturer's instructions.

The extracted DNA was enriched for mtDNA, using exonuclease V, which selectively digests linear DNA while preserving the circular mtDNA (*80*), as described previously (*26*, *27*). This step was necessary to minimize the contribution from Numts (*81*). In short, ~2.5 µg of total DNA was incubated at 37°C for 48 hours in 60-µl reactions containing 1× NEB 4 buffer, 30 U Exonuclease V (RecBCD, NEB), and 2 mM ATP. After this first incubation, the reaction volume was increased to 100 µl by supplementing with further NEB 4 buffer (final concentration = 1×), 20 U Exonuclease V, ATP (final concentration = 2 mM), and 5 µg of RNase A (Amresco) and incubated at 37°C for another 16 hours. The reactions were purified using 1 volume of AMPure XP beads (Beckman Coulter).

Oocyte lysis and mtDNA enrichment

Oocytes were lysed and mtDNA enriched as previously described (*26*, *27*). In short, 4 µl of cold oocyte lysis buffer [50 mM tris-HCl (pH 8.0; Invitrogen), 1 mM EDTA (pH 8.0; Promega), 0.5% polyoxyethylene (20) sorbitan monolaurate (OmniPur), and Proteinase K (0.2 mg/ml; IBI Scientific)] were added to a single oocyte in a low-binding tube, incubated at 50°C for 15 min, followed by an overnight incubation at 37°C. Proteinase K was then partially inactivated by incubating at 65°C for 15 min. A solution of 1 mM ATP, 10 mM $MgCl_2$, 10 mM tris-HCl (pH 8.0), and 10 U Exonuclease V (RecBCD, NEB) was added to obtain a final volume of 13 µl, followed by incubation at 37°C for 1 hour. Ten nanograms of RNase A (Amresco) was added, and reactions were incubated at 37°C for 15 min. Tris-EDTA (TE) buffer was added up to the final volume of 50 µl. Enzymes were inactivated by incubation of the samples at 70°C for 30 min.

mtDNA enrichment estimation

mtDNA enrichment was estimated by real-time PCR, amplifying a region of the mitochondrial *ND6* gene, as well as nuclear *Alu* elements. Samples were amplified using primers specific for mtDNA [F-hsND6-mtDNA: CCACAGCACCAATCCTACCT and R-hsND6-mtDNA: GTCAGGGGTTGAGGTCTTGG (both IDT); NC_012920.1: positions 14,259 to 14,525] as well as for nuclear DNA [F-primateAlu: CCTGTAATCCCAGCACTTTG and R-primateAlu: TAGTAGAGACGGGGTYTCAC (both IDT), amplifies multiple regions in the genome]. Reactions contained 2 µl of 1:10 diluted ligation reaction, 1× PowerUp SYBR Green Master Mix (Thermo Fisher Scientific), and 0.2 µM each primer and were amplified at 95°C for 2 min, 45 cycles of 95°C for 15 s, 57°C for 20 s, and 72°C for 30 s, followed by 72°C for 2 min and a melting curve analysis. A standard consisting of tissue samples that were previously enriched for mtDNA at nine different efficiencies (1.8, 11.5, 17.5, 30.0, 33.1, 59.5, 61.6, 69.2, and 93.0% of mtDNA in the total sample as determined using next-generation sequencing) was used to estimate the mtDNA enrichment.

Duplex library preparation and sequencing

Duplex libraries were prepared as described previously (*26*, *27*). In short, the enriched DNA was sheared with a Covaris M220 Focused-ultrasonicator using 50-µl reactions in microTUBE-50 AFA Fiber

Screw-Cap (Covaris), using the protocol for ~550–base pair (bp) fragments. Samples were transferred into 200-µl low-binding tubes (Axygen) for library preparation. Duplex sequencing libraries were prepared following a previously described protocol (*82*) with several modifications, which are provided in more detail in a prior study (*27*). In short, T-tailed duplex adapters were prepared with sequences MWS51 and MWS55. End-repair and A-tailing were performed with the End Prep module of the NEBNext Ultra II DNA Library Prep Kit for Illumina (NEB) according to the manufacturer's instructions, followed by adapter ligation with the NEBNext Ultra II Ligation Mix and NEBNext Ligation Enhancer, using ~25 fmol of duplex adapters for somatic tissues and ~18 fmol for oocytes. Adapter-ligated DNA was purified twice with 0.8 volumes of AMPure XP beads (Beckman Coulter) and eluted in 15 µl of 10 mM tris-HCl (pH 8.0). The amount of amplifiable fragments and hence the input for tag-family PCR was estimated with real-time PCR: 2 µl of the 1:10 diluted adapter-ligated library aliquot was amplified in 10-µl reactions containing 1 µM NEBNext Universal PCR Primer for Illumina (NEB), 1 µM NEB_mws20 primer [GTGACTGGAGTTCAGACGTGTGCTCTTCCGATC*T (IDT); the asterisk indicates a phosphorothioate bond], 1× Kapa HiFi HotStart Reaction Mix (KAPA Biosystems), and 1× EvaGreen Dye (Biotium). Reactions were amplified in a CFX96 Real-Time PCR machine (Bio-Rad) at 98°C for 45 s, 35 cycles of 98°C for 15 s, 65°C for 30 s, and 72°C for 45 s, followed by 72°C for 2 min and a melting curve analysis. PCR products were additionally inspected on a 1.5% agarose gel. Tag-family PCR reactions contained up to 14 µl of the adapter-ligated sample. If the Cq value in the real-time PCR was ≥23, all 14 µl of the sample was used as PCR template, and samples with lower Cq values were diluted accordingly. For samples with a real-time PCR Cq value of ~23, 7M paired-end reads were anticipated during sequencing, ensuring efficient DCS formation. For tag-family PCR, first, 12 cycles of linear amplification were performed in 38-µl reactions: 14 µl (accordingly diluted) of DNA, 1× Kapa HiFi HotStart Reaction Mix (KAPA Biosystems), and 1 µM NEB_mws20 primer at 98°C for 45 s, 12 cycles of 98°C for 15 s, 60°C for 30 s, and 72°C for 45 s, followed by 72°C for 2 min. A 1 µM solution of NEBNext Universal PCR Primer for Illumina (NEB) was added to each reaction followed by amplification at 98°C for 45 s, nine cycles of 98°C for 15 s, 65°C for 30 s, and 72°C for 45 s, followed by 72°C for 2 min. PCR products were purified with 0.8 volumes of AMPure XP beads (Beckman Coulter) using the epMotion automated liquid handling system (Eppendorf), and different indexes were added via PCR: In 50-µl reactions, 15 µl of purified library was amplified with 1 µM NEBNext Universal PCR Primer for Illumina (NEB), 1 µM NEBNext Multiplex Oligos for Illumina (Index Primers Sets 1 to 4), and 1× Kapa HiFi HotStart Reaction Mix (KAPA Biosystems) at 98°C for 45 s, 9 to 16 cycles of 98°C for 15 s, 65°C for 30 s, and 72°C for 45 s, followed by 72°C for 2 min. The number of required cycles was estimated based on the amount of sample used as input in the linear PCR. PCR products were purified with 0.8 volumes of AMPure XP beads (Beckman Coulter) using the epMotion automated liquid handling system (Eppendorf), their quality checked with a Bioanalyzer High Sensitivity DNA Kit (Agilent), quantified with the KAPA Library Quantification Kit (Kapa Biosystems) according to the manufacturer's instructions, and pooled depending on the number of reads needed per sample to obtain a family size of around 6. Sequencing was performed on an Illumina HiSeq 2500 platform with 250-nt paired-end reads.

Targeted mtDNA capture sequencing

Because mtDNA enrichment with exonuclease V turned out to be inefficient for low mtDNA copy number samples such as blood and saliva, mtDNA in somatic tissues was also enriched using the targeted mtDNA capture approach, as previously described for duplex sequencing (*83*). This was done for the 18 blood and 11 saliva samples (see table S1 and the section "Experimental design, study participant details, samples, and their handling" for more details). Duplex libraries were prepared as described above (with minor modifications), starting with 500 ng of total DNA in a volume of 50 µl, sheared to ~550 bp with a Covaris M220 Focused-ultrasonicator. Modifications included the use of ~25 pmol of duplex adapters and 120 ng of adapter-ligated DNA as input for tag-family PCR (run for three cycles; no linear amplification).

Targeted mtDNA capture was performed using the xGen Hybridization and Wash Kit and the xGen Human mtDNA Research Panel v1.0 (both IDT) according to the manufacturer's instructions, with minor modifications for compatibility with the duplex sequencing protocol. Modification included the use of the previously published mws60 and mws61 blocking oligos (*83*) and the post–capture (indexing) PCR that was performed in 60-µl reactions as follows: 20 µl of purified library was amplified with 1 µM NEBNext Universal PCR Primer for Illumina (NEB), 1 µM NEBNext Multiplex Oligos for Illumina (Index Primers Sets 1 to 4), and 1× Kapa HiFi HotStart Reaction Mix (KAPA Biosystems) at 98°C for 45 s, 15 cycles of 98°C for 15 s, 65°C for 30 s, and 72°C for 30 s, followed by 72°C for 2 min. For each sample, ~9 M paired-end reads were anticipated during sequencing. Sequencing was performed on an Illumina HiSeq 2500 platform with 250-nt paired-end reads.

Consensus read formation, mapping, and variant calling

Single-strand (SSCS) and duplex (DCS) consensus formation was performed in Galaxy (*84*) using the Du Novo pipeline (version 2.14) including barcode error correction (*28*, *29*), as described previously (*26*, *27*), and the quality of the sequencing reads was verified using FastQC. In short, barcode error correction allowed ≤3 mismatches and required a mapping quality of ≥20; a family size of ≥3 was required for consensus formation, calling a mutation if present in ≥70% of the sequencing reads. To avoid errors associated with end preparation, the first 12 nucleotides of the consensus sequences were trimmed using trimmomatic (HEADCROP) (*85*). Trimmed DCSs and SSCSs were mapped to the human reference genome (hg38, which contains the rCRS mtDNA reference sequence NC_012920.1) using BWA-MEM (*86*).

Fixed differences between individual donors were used to check for potential cross-sample contamination during library preparation. When comparing libraries prepared in the same batches, no evidence of cross-sample contamination was found. mtDNA enrichment efficiency was estimated by calculating the percentage of sequences mapping to the mtDNA reference. To minimize mapping bias at the edges from mapping a circular genome to a linearized reference, in addition to mapping to the whole mtDNA reference (as part of hg38), reads were mapped to hg38 in which chrM was replaced by a relinearized reference sequence (consisting of positions 14,570 to 16,569 and 1 to 2000). Positions 16,070 to 500 were analyzed from the sequences mapped to the relinearized mtDNA reference version. BAM files were filtered using BAMTools (*87*), requiring a mapping quality of >20, mapping to chrM, being the primary alignment,

paired reads, proper read pairing, and mapping of the mate sequence. Reads were left aligned using the Bam Left Align tool.

To avoid a bias in variant frequencies from overlapping paired-end reads, overlaps were clipped using the clipOverlap function of bamUtil (*88*). Variants were called using the naive variant caller (*89*) with the following settings: a minimum number of one read needed to be considered as REF/ALT, minimum base quality of 20, minimum mapping quality of 50, and ploidy of 1. DCSs were used for the initial variant calling and for de novo mutation analysis, whereas SSCSs were used in addition to DCSs for the analysis of inheritable heteroplasmies (see corresponding section for more details).

Variant filtering and classification of tissue-specific de novo mutations

We identified 4643 variants; for 11 of these, two different mutant alleles were recorded; however, by default only one allele was displayed (the variant of the second mutant allele was added manually). We excluded from our analyses 468 variants, located at the following 15 mtDNA positions: 302, 310, 312, 315 to 317, 513, 514, 515, 523, 525, 3107, 16,183, 16,189, and 16,194. These are mainly in repetitive sequences, and the excluded variants were identified as false-positive mutations occurring at the beginning or at the end of mononucleotide repeats due to mapping errors. In addition, we excluded 71 donor-specific variants resulting from repetitive sequences at the following positions: hs022 545 G>C, hs022 564 G>A, hs022 567 A>C, hs022 574 A>C, hs022 8281 C>A, hs004 8279 T>C, hs015 8279 T>C, hs022 8285 C>T, hs004 8281 C>G, hs015 8281 C>G, hs011 16186 T>C, hs015 5895 C>A, hs015 5899 C>A, hs003 16180 A>C, hs005 16180 A>C, hs015 16180 A>C, hs005 16181 A>C, hs005 16181 C>A, and hs005 16182 C>A. We also excluded 222 linked variants (with >3 variants present in single paired-end reads, mostly occurring in multiple unrelated samples), which represent potential Numts. Among the remaining variants, we further excluded 146 variants with ambiguous timing of occurrence during development (for these, we could not unambiguously determine whether they represented inheritable heteroplasmies, early somatic mutations, or early germline mutations). A total of 148 inheritable heteroplasmies [i.e., variants present in all somatic tissues assessed and ≥1 oocyte(s) of a donor] were analyzed separately, as described in more detail in the section "Analysis of inheritable heteroplasmies." Last, 63 low-confidence variants identified in oocytes with a mean DCS depth of <50× (originating from two oocytes) were excluded because the computation of mutation frequencies could be biased by the low number of total nucleotides sequenced for each oocyte. As a result, we obtained 3525 high-confidence tissue-specific de novo mutations. Mutations measured in >1 DCS per sample were counted as a single mutation, assuming that they resulted from a single mutation event. Mutation frequency was computed as the number of mutations divided by the product of mtDNA length (16,569 bp) and mean DCS depth across samples.

Analysis of de novo mutations

Generalized mixed-effects linear model to assess the effect of age

The effect of age on mutations was assessed using a generalized mixed-effects linear model, as described previously (*27*). In short, for each sample, we computed the mutation frequency as the number of observed mutations divided by the total number of sequenced nucleotides and used the R function glmer (from the package lme4) to fit a generalized mixed-effects linear model with binomial family and logit link, the mutation frequency as response and the numbers of sequenced nucleotides as weights (prior weights are used to provide the number of trials in a binomial model when the response is the proportion of successes). We included age, the categorical variable tissue, and their interaction as fixed-effect predictors, whereas donor ID was used as a random effect. The variable age was standardized before fitting the model to avoid convergence issues. The Wald test—as provided by the summary of the fitted glmer model—was used to ascertain whether each of the coefficients was null. Age effect, tissue effect, and random effects were tested using the likelihood ratio test based on the chi-square distribution provided by the function anova.merMod. The conditional modes of the random effects were extracted using the function ranef. We also tested the effect of age for each of the tissues considered using the Z test provided by the function glht of the R package multcomp. Marginal and conditional pseudo-R^2 (representing the variability explained by the fixed effects and by the entire model, respectively) were computed using the function r.squaredGLMM (from the R package MuMIn). Predictions of mutation frequencies based on the fixed-effect part of the model were computed using the function predict.merMod from the R package lme4, whereas bootstrap pointwise CIs based on the quantile approach were computed using the function bootMer (R package lme4).

Estimation of the rate of nonsynonymous versus synonymous mutations

We tested for selection in protein-coding sequences by calculating hN/hS ratios, as described in (*32*). hN is the number of nonsynonymous variants per nonsynonymous sites, and hS is the number of synonymous variants per synonymous sites. The numbers of nonsynonymous and synonymous sites were calculated using the Nei-Gojobori method (*90*). Under neutrality, the hN/hS ratio is expected to be equal to 1; a ratio of <1 is suggestive of purifying selection, and a ratio of >1 is suggestive of positive selection. The neutral distribution of hN/hS was generated by bootstrapping (*32*) (100 replicates) from the observed mutation spectrum and used to assess whether the observed hN/hS ratio deviates from neutral expectations.

Conservation analysis

Analogous to previous studies (*26*, *40*), PhastCons 100way scores for human mtDNA (hg38) were downloaded from the UCSC genome browser (*91*), and selection acting on all mutations (including those occurring in noncoding regions) was assessed as follows: We first defined sites with PhastCons scores > 0.75 as conserved, sites with scores < 0.25 as nonconserved, and sites in the reference genome as mutated or nonmutated (depending on the analyzed mutations). Mean and median PhastCons scores were calculated separately for mutated or nonmutated sites, either for the whole mtDNA or separately for D-loop and coding region.

A Fisher's exact test was performed to test whether mutations were more or less likely to occur at nonconserved sites than at conserved sites, and the odds ratio (odds of observing a mutation at nonconserved site/odds of observing a mutation at a conserved site) was calculated. Significance of odds ratio deviation from 1 was tested nonparametrically using a permutation test: We "rotated" the mtDNA genome by a random number and tested the association between conservation status and mutation status and compared the observed data to the permuted data (*26*).

Assessment of local constraints

We used the recently published MLC score (*43*) to assess differences in functional consequences of variants. MLC scores for human

mtDNA were downloaded from (*43*) and classified into disease-associated variants (see "Annotation of disease-associated variants" below) and variants without known disease association.

Identification of variant hotspots

Variant hotspots were identified as described previously (*27*). We define variant hotspots as mutations that are shared by multiple donors but are not expected to be shared by chance (168 mutations were shared in oocytes and 161 mutations were shared in somatic tissues). Blood and saliva were considered together as "somatic tissues" due to similar mutation frequencies and lower numbers of measured mutations, hence increasing our power to detect somatic variant hotspots. For the evaluation of oocyte variant hotspots, we also considered the number of sequenced oocytes per donor (ranging from 1 to 5).

In short, we built a probabilistic model that takes into account tissue-specific estimates of the average mutation frequencies inside and outside of the D-loop, as well as the mean DCS depth in each sample. We computed the expected number of mutations present in exactly one woman and shared by several women, separately for oocytes and somatic tissues. As a result, we defined oocyte variant hotspots as sites mutated in oocytes of at least four different women and somatic hotspots as sites mutated in somatic tissues of at least three different women.

Annotation of disease-associated variants

Mutations known to be associated with diseases were annotated based on the 109 disease-associated confirmed substitutions reported by MITOMAP (*33*) (downloaded on 18 June 2024). They were located in protein-coding regions ($n = 56$), tRNA ($n = 51$), or rRNA ($n = 2$).

Analysis of inheritable heteroplasmies

Identification of inheritable heteroplasmies

All sequenced oocytes and somatic tissues were included in the analysis of inheritable heteroplasmies. MAFs obtained from variant calling on DCSs were primarily used in the analysis of inherited heteroplasmies. However, if the sequencing depth for a minor allele was <3 for DCSs, the MAFs obtained from the SSCS analysis were also used, as described previously (*26*, *27*). For somatic tissues, in addition to exonuclease V mtDNA enriched sequencing libraries, we included duplex libraries enriched for mtDNA using a targeted capture approach. Although such samples harbor the chance of Numt coenrichment, the sole assessment of variants previously identified in donors using exonuclease V mtDNA enrichment diminishes such bias. This increased the average SSCS sequencing depth from 445× and 514× to 3750× ($n = 18$, SD = 824.6) and 5019× ($n = 11$, SD = 1864.1) per site per sample in blood and saliva, respectively. MAFs measured using the two different enrichment strategies correlated well (fig. S9).

mtDNA bottleneck calculation

The size of the effective germline mtDNA bottleneck was estimated as described previously (*27*). In short, the effective mtDNA bottleneck N was calculated by first computing N_{ij} for individual i and position j as

$$N_{ij} = \frac{p_{ij}(1-p_{ij})}{\sigma^2_{\text{oocyte}}}$$

where p_{ij} is the mean allele frequency of blood and saliva (if both sample types were available), and the denominator is the mean square deviation (across n_{ij} oocytes for the same donor and position) calculated as

$$\sigma^2_{\text{oocyte}} = \frac{\sum_{k}^{n_{ij}} \left(p_{ijk}^{\text{oocyte}} - p_{ij}\right)^2}{n_{ij}}$$

where p_{ijk}^{oocyte} is the allele frequency for the kth oocyte. To obtain N, we averaged the N_{ij}'s. The 95% CIs were generated by bootstrapping (sampling with replacement 1000 times).

Statistical analysis: Significance tests

One-sided permutation tests were performed to assess whether the median mutation frequencies were higher in older age groups for each of the tissues analyzed, using 10,000 permutations. Larger expected quantities in older donors justify the use of a one-sided test. Significance of age-related increases after combining all mutations across samples belonging to the same group (e.g., mutations in blood of young women) was assessed using the Fisher's exact test (function fisher.test from the stats package in R). P values were corrected for multiple comparisons (function p.adjust from the stats package in R) using the method of Benjamini, Hochberg, and Yekutieli to control the false discovery rate (*92*).

Supplementary Materials

This PDF file includes:

Figs. S1 to S13
Tables S1 to S19

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under accession number PRJNA1181658. The data were analyzed in R using the packages listed above. Tables with all mutations were uploaded to GitHub, available at https://github.com/makovalab-psu/human_mtDNA_duplexSeq, as well as Zenodo under DOI: 10.5281/zenodo.15550930.

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